Objective:

To perform risk analysis and calculate hedging parameters on a portfolio of 2-year and 10-year U.S. Treasury notes using a two-year window of daily yield shock data.

List of Questions Included:

1. For each "shock" series: What percent of the daily changes are within 1 standard deviation of the corresponding mean change? How about within 2 standard deviations?
2. For the complete 2-year window: What is the sample correlation?
3. Compute these conditional expected values of the sample:
   - E[2-year shock | 10-year shock >= 0]
   - E[10-year shock | 2-year shock >= 0]
   - Explain how these statistics carry their own information on the correlation.
4. Compute the sample correlation restricted to the subset of the days for which, for each series, the change is greater than 1 standard deviation away from the mean (for the 2-year window). How many days are in this subset?
5. Suppose we are long $20 million 10-year notes. How many 2-year notes should we sell so that the composite position is PV01-neutral?
6. Suppose we are long $15 million 2-year notes and long the required number of 10-year notes for which each component contributes the same amount of PV01. Does this imply that each component contributes the same "amount" of risk?
7. For the portfolio described in [6], $15 million of 2-year notes and $3.816 million of 10-year notes, compute the 99% 1-day VaR. Please use the second-order price/yield repricing formula in the spreadsheet.
8. Let P(A) denote the portfolio holding $15 million 2-year notes and A million 10-year notes. Generate a graph showing the 99% 1-day VaR of P(A) as a function of A. Is the A that obtains the lowest VaR the value corresponding to the zero total PV01? Can you comment?